Electromagnetic induction

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Explain why changing magnetic flux can induce an emf.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 11.*
E=Pt=15 J.

02. Worked example: explain the model
Motion changes flux; stationary unchanging flux gives zero induced emf.

03. Read and interpret the model
Moving magnet can change flux; the stationary arrangement does not produce sustained emf from motion.

04. Apply the physics
Opposite sign, for unchanged coil and meter connections.

05. Explain your reasoning
If external flux decreases, the induced field can reinforce its original direction to oppose the decrease.

06. Review: connect an earlier idea *Review: Unit 7, Day 2.*
Force magnitude becomes 4 times as large, from Coulomb’s law.

07. Exit check
Move faster, increase turns, increase field change or effective area; accept with relevant controls.

08. STRETCH: test the model
Greater rate of flux change or more identical turns can increase voltage; hold geometry/field/path or speed fixed for fair comparison.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 11.*
V=15 V; $P=IV=I^2R=75$ W.

02. Worked example: explain the model
Motion changes flux; stationary unchanging flux gives zero induced emf.

03. Read and interpret the model
Moving magnet can change flux; the stationary arrangement does not produce sustained emf from motion.

04. Apply the physics
0.80 V.

05. Explain your reasoning
If external flux decreases, the induced field can reinforce its original direction to oppose the decrease.

06. Review: connect an earlier idea *Review: Unit 7, Day 2.*
$F_f/F_i=16/36\approx0.4444$.

07. Exit check
Four times; same number of turns and same flux change.

08. STRETCH: test the model
7.5 V; induced effect opposes the change in flux, not necessarily the existing field itself.